Path Codes on Shared Knowledge Graphs

Part 1: The Price of Memory

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Abstract

A path code conveys a meaning by describing a walk on a shared graph, decision by decision. We compare three ways of doing so. *Native* codes use only the shared routing statistics and pay, above the semantic entropy $H(M)$, the average conditional min-entropy of the walk given the meaning. *Adapted* codes, which also share the selection of one walk per meaning, attain $H(M)$. They need no decoder memory when the meaning is determined by the endpoint of the walk, but may need it when the meaning depends on the route (context-dependent meaning, or polysemy). *Bits-back* codes attain $H(M)$ with native statistics, at the price of coupling consecutive messages. A corollary quantifies when the semantic Markov sources of Xu et al. [3] have the entropy their formula assigns to them.

1 Setting

We use the model of Part 0: a trajectory network G satisfying Assumptions 1–3, path measure P, meaning map $\mu : \Pi_G \to \mathcal{M}$, preimages $\Pi(m)$, and meaning law P_M. Code lengths are *ideal*: a sequential code that uses distribution Q assigns $-\log_2 Q(\pi)$ bits to walk π. By Lemma 1 of Part 0, arithmetic coding realises any such length within 2 bits [7, 8], so every statement below holds for real codes up to an additive 2.

Definition 1 (Selection and induced law)**.** A *selection* is a map $f : \mathcal{M} \to \Pi_G$ with $f(m) \in \Pi(m)$. It induces the law Q_f on walks given by $Q_f(f(m)) = P_M(m)$ and $Q_f(\pi) = 0$ for walks outside the image of f.

Definition 2 (Native, adapted and bits-back path codes)**.**

1. A *native* code with selection f describes $f(m)$ using the shared routing kernel, at cost $-\log_2 P(f(m))$. Its rate is $L_{\mathrm{nat}}(f) = \mathbb{E}[-\log_2 P(f(M))]$. Transmitter and receiver share only G and μ.

2. An *adapted* code with selection f and state space S is a finite-state decoder. When the walk so far is ρ and ends at node v, the decoder is in state $s(\rho) \in S$. It decodes the next decision with a kernel $q(\cdot \mid v, s)$, and moves to $s(\rho e) = \delta(s(\rho), e)$. Its rate is $L_{\mathrm{ad}} = \mathbb{E}_{Q_f}[-\log_2 q(\Pi)]$, where $q(\pi)$ is the product of the kernels along π. It is *memoryless* if $|S| = 1$. In addition to G and μ, both ends share f, q and δ.

3. A *bits-back* code uses the native kernel and a randomised selection $\pi \sim P(\cdot \mid m)$, and recovers the selection bits at the receiver [5, 6].

Every path code lets the receiver recover the meaning, so its rate is at least $H(M)$, the semantic entropy H_s of [1].

2 Native codes

For $m \in \mathcal{M}$ let $\pi^\star(m) \in \arg\max_{\pi \in \Pi(m)} P(\pi)$, with ties broken by a fixed order, and define

$$g(m) = \log_2 \frac{P_M(m)}{P(\pi^\star(m))} = -\log_2 \max_{\pi \in \Pi(m)} P(\pi \mid M = m). \tag{1}$$

Thus $g(m)$ is the min-entropy of the walk given the meaning, $H_\infty(\Pi \mid M = m)$.

Theorem 1 (Price of native routing)**.** *(a)* $\min_f L_{\mathrm{nat}}(f) = L_{\mathrm{nat}}(\pi^\star) = H(M) + \mathbb{E}[g(M)]$.

(b) For every m, $0 \le g(m) \le H(\Pi \mid M = m) \le \log_2 |\Pi(m)|$. *Hence* $0 \le \mathbb{E}[g(M)] \le H(\Pi \mid M)$.

(c) $\mathbb{E}[g(M)] = 0$ *if and only if every meaning has exactly one walk of positive probability.* $\mathbb{E}[g(M)] = H(\Pi \mid M)$ *if and only if, for every* m *with* $P_M(m) > 0$, *the conditional law* $P(\cdot \mid M = m)$ *is uniform on its support.*

Proof. (a) The cost of a native code is $\sum_m P_M(m)\,(-\log_2 P(f(m)))$, which is minimised term by term by $f(m) = \pi^\star(m)$. Writing $-\log_2 P(\pi^\star(m)) = -\log_2 P_M(m) + g(m)$ and averaging gives the claim.

(b) $g(m) \ge 0$ because $P(\pi^\star(m)) \le P_M(m)$. Fix m and write $c(\pi) = P(\pi \mid M = m)$. Then $g(m) = -\log_2 \max_\pi c(\pi) \le \sum_\pi c(\pi)(-\log_2 c(\pi)) = H(\Pi \mid M = m)$, because $-\log_2 \max_\pi c(\pi) \le -\log_2 c(\pi)$ for every π. The last inequality is the usual bound of entropy by the logarithm of the support size [9]. Averaging over M gives the second statement.

(c) $\mathbb{E}[g(M)] = 0$ if and only if $g(m) = 0$ for every m with $P_M(m) > 0$, that is, $\max_\pi c(\pi) = 1$. For the upper case, $\mathbb{E}[g(M)] = H(\Pi \mid M)$ requires equality in (b) for every such m. The inequality $-\log_2 \max c \le -\log_2 c(\pi)$ is tight in average if and only if $c(\pi) = \max c$ on the support, that is, c is uniform there. □

Remark 1 (Kraft slack). Native lengths waste coding space:

$$\sum_m 2^{\log_2 P(\pi^\star(m))} = \sum_m P(\pi^\star(m)) = 1 - \sum_m \sum_{\pi \in \Pi(m) \setminus \{\pi^\star(m)\}} P(\pi).$$

The mass of the unused synonymous walks is coding space that no message ever uses.

3 Adapted codes and decoder memory

For a selection f and a prefix ρ of some walk in the image of f, let $Q_f(\cdot \mid \rho)$ be the law of the remaining decisions, given that the selected walk starts with ρ. For a node v, let

$$N_v(f) = \big|\{Q_f(\cdot \mid \rho) : \rho \text{ ends at } v,\ Q_f(\rho) > 0\}\big|$$

be the number of distinct future laws at v, and $N(f) = \max_v N_v(f)$.

Theorem 2 (Adapted codes)**.** *(a) For every selection* f *there is an adapted code with* $L_{\mathrm{ad}} = H(M)$.

(b) An adapted code with selection f *attains* $L_{\mathrm{ad}} = H(M)$ *if and only if its state assignment separates prefixes that end at the same node and have different future laws. The least number of states that suffices is* $N(f)$. *Any code that merges two such prefixes has* $L_{\mathrm{ad}} > H(M)$.

(c) If the meaning is endpoint-determined *($\mu(\pi)$ depends only on the last node of π), then the Viterbi selection $\pi^\star$ satisfies $N(\pi^\star) = 1$, so a memoryless adapted code attains $H(M)$.*

(d) There exist networks with route-dependent meaning for which $N(f) \geq 2$ for every selection f. For them, every memoryless adapted code has $L_{\mathrm{ad}} > H(M)$.

Proof. (a) Since f is injective, Q_f is P_M transported to the image of f, so $H(Q_f) = H(M)$. Take S to be the set of prefixes and $q(\cdot \mid v, \rho) = Q_f(\cdot \mid \rho)$. The kernels then multiply to $Q_f(\pi)$ along each selected walk, and the rate is $\mathbb{E}_{Q_f}[-\log_2 Q_f(\Pi)] = H(Q_f) = H(M)$.

(b) For any adapted code, $L_{\mathrm{ad}} = \mathbb{E}_{Q_f}[-\log_2 q(\Pi)] = H(Q_f) + D(Q_f \| q)$, where q denotes the walk law the code induces. By the chain rule for relative entropy,

$$D(Q_f \| q) = \sum_{\rho} Q_f(\rho)\, D\big(Q_f(\cdot \mid \rho)_1 \,\big\|\, q(\cdot \mid v(\rho), s(\rho))\big),$$

where the sum runs over prefixes, $Q_f(\cdot \mid \rho)_1$ is the law of the next decision, and $v(\rho)$ is the last node of ρ. The rate equals $H(M)$ if and only if every term vanishes, i.e. the kernel at $(v(\rho), s(\rho))$ equals the next-decision law of every prefix assigned to that state. Two prefixes ending at v with different future laws differ in the next-decision law at some later prefix along a common continuation, so a state map that merges them makes some term positive. Conversely, the map "prefix $\mapsto$ its future law" is a valid state assignment: the future law of ρe is the future law of ρ conditioned on the first decision being e, hence a function of the future law of ρ and of e. This defines δ, and uses exactly $N_v(f)$ states at v.

(c) Define $\rho^\star(r) = (r)$ and, in a topological order of the DAG, $\rho^\star(v) = \rho^\star(u)\, v$ with u a predecessor maximising $P(\rho^\star(u))p(v \mid u)$, ties broken by a fixed order. By induction, $\rho^\star(v)$ is a most probable walk from r to v, and the set $\{\rho^\star(v)\}_v$ is prefix-closed, i.e. it forms a tree rooted at r. Since the meaning is endpoint-determined, $\max_{\pi \in \Pi(m)} P(\pi) = \max_{t:\, \mu(t)=m} P(\rho^\star(t))$. Choosing for each m a maximising terminal t_m and setting $\pi^\star(m) = \rho^\star(t_m)$ gives a Viterbi selection whose walks are branches of that tree. Every node is then reached by at most one selected prefix, so $N_v(\pi^\star) \leq 1$ for every v.

(d) Example 2 below has $N(f) = 2$ for each of its selections. The strict inequality then follows from (b). □

Remark 2 (What memory buys). Theorems 1 and 2 separate three resources. The native code needs only the shared graph and pays $\mathbb{E}[g(M)]$. The adapted code removes that payment but requires sharing a selection; it never requires decoder memory when the meaning is endpoint-determined, and it can require it when the meaning of a decision depends on how the walk arrived there. In the language of the original note, *polysemy forces the walker to carry state*: Theorem 2(c)–(d) makes this precise in one direction and shows by example that the other can fail. The least memory $\min_f N(f)$ is a combinatorial quantity whose computational complexity is left open.

4 Bits-back codes

Proposition 1 (Native statistics, amortised optimality)**.** *Under the standard idealised model of bits-back coding [5, 6], the expected total length to send n i.i.d. meanings is at most $nH(M) + H(\Pi \mid M) + 2$. The rate therefore tends to $H(M)$ using only G and μ.*

Proof sketch. For each message the encoder draws $\pi \sim P(\cdot \mid m)$ by *decoding* from a stack of previously produced bits, which consumes $-\log_2 P(\pi \mid m)$ bits on average. It then encodes π with the native kernel, which adds $-\log_2 P(\pi)$ bits. The receiver decodes π, computes $m = \mu(\pi)$,

and re-encodes π under $P(\cdot \mid m)$, which restores the consumed bits. The net cost per message is $-\log_2 P(\pi) + \log_2 P(\pi \mid m) = -\log_2 P_M(m)$, whose mean is $H(M)$. Only the first message needs fresh bits, $H(\Pi \mid M)$ on average. The final arithmetic-coding flush adds at most 2 bits. □

5 Stationary sources: when is a semantic Markov chain Markov?

Let (X_k) be a stationary Markov chain on a finite set, φ a synonymous mapping of its states, and $Y_k = \varphi(X_k)$. Xu et al. [3, Def. 1] define the semantic transition matrix by stationary-weighted aggregation. Under stationarity this is exactly the law of Y_2 given Y_1. Their block entropy [3, Thm. 3], for $m = 1$, is

$$\widehat{H}_N = H(Y_1) + (N-1)\, H(Y_2 \mid Y_1).$$

Corollary 1 (First-order semantic models)**.** *(a) The excess of the first-order formula is*

$$\widehat{H}_N - H(Y_1, \ldots, Y_N) = \sum_{k=3}^{N} I\big(Y_k; Y_1^{k-2} \mid Y_{k-1}\big) \geq 0.$$

(b) $\widehat{H}_N = H(Y_1, \ldots, Y_N)$ for all N if and only if (Y_k) is a Markov chain. This holds, for instance, when X is lumpable with respect to φ [4].

(c) The per-symbol excess $\lim_N (\widehat{H}_N - H(Y_1^N))/N = H(Y_2 \mid Y_1) - \bar{H}(Y)$ is the price of a memoryless semantic model, where $\bar{H}(Y)$ is the entropy rate.

Proof. By the chain rule, $H(Y_1^N) = H(Y_1) + \sum_{k=2}^{N} H(Y_k \mid Y_1^{k-1})$. By stationarity, $H(Y_k \mid Y_{k-1}) = H(Y_2 \mid Y_1)$. Hence

$$\widehat{H}_N - H(Y_1^N) = \sum_{k=2}^{N} \big[H(Y_k \mid Y_{k-1}) - H(Y_k \mid Y_1^{k-1})\big] = \sum_{k=3}^{N} I(Y_k; Y_1^{k-2} \mid Y_{k-1}),$$

where the $k = 2$ term vanishes. This proves (a). Each term is zero if and only if Y_k is conditionally independent of Y_1^{k-2} given Y_{k-1}; all of them vanish if and only if (Y_k) is Markov, which proves (b). For (c), divide by N and use $H(Y_k \mid Y_1^{k-1}) \to \bar{H}(Y)$. □

Remark 3. In words: the block entropy of [3, Thm. 3] is exact when the aggregated process is Markov, and in general it is the code length of a first-order model of the semantic source. In 200 random four-state chains with the partition $\{1\}, \{2\}, \{3, 4\}$ and $N = 6$, the excess reached 1.11 bits (script `check_xu.py`). The excess plays here the role that $\mathbb{E}[g(M)]$ plays for walks: it is what a decoder without memory pays.

6 Examples

Example 1 (Endpoint-determined meaning: Part 0 revisited). In the "what day is today" network the meaning is endpoint-determined (Tuesday and Tuesday$'$ are distinct terminals). Theorem 1 gives the native rate $H(M) + \mathbb{E}[g(M)] = 1.485 + 0.737 = 2.222$ bits, with $\mathbb{E}[g(M)] = 0.737 < H(\Pi \mid M) = 0.971$. The bound is not tight because $P(\cdot \mid M = m) = (0.6, 0.4)$ is not uniform. By Theorem 2(c), the Viterbi selection routes everything through "today is", and a memoryless adapted kernel attains $H(M) = 1.485$ bits.

Example 2 (Context-dependent meaning forces memory). Let $r \to a$ with probability 0.6 and $r \to b$ with probability 0.4. Both a and b lead to a shared node x, a "word". From x the walk ends at t_1, t_2, t_3 with probabilities $(0.5, 0.3, 0.2)$. The meaning depends on the context branch:

walk	$a\,t_1$	$a\,t_2$	$a\,t_3$	$b\,t_1$	$b\,t_2$	$b\,t_3$
P	0.30	0.18	0.12	0.20	0.12	0.08
meaning	A	A	B	A	C	C

The word x is polysemous: t_2 means A after a and C after b. Here $P_M = (0.68, 0.12, 0.20)$, $H(M) = 1.210$ and $H(\Pi \mid M) = 1.247$ bits.

- *Native code:* 2.160 bits, an excess of 0.950 bits.
- *Any selection needs two states at x.* The future law at x after a (resp. b) is the law of the endpoints of the selected walks through a (resp. b). B has the single walk $a\,t_3$, and C only walks through b. If A is selected through a, the support after a has two endpoints and the support after b has one. If A is selected through b, the support after a is $\{t_3\}$ and after b has two endpoints. In both cases the two laws differ, so $N(f) = 2$ for all six selections $(3 \times 1 \times 2)$, as an exhaustive check confirms.
- *Memoryless adapted codes:* the best one over all selections and node kernels has rate 1.251 bits, an excess of 0.041 bits. With the Viterbi selection the excess is 0.722 bits.
- *Adapted code with two states at x:* rate $H(M) = 1.210$ bits.

Script: `contraejemplo_memoria.py`.

7 What a reviewer will ask (and the answers)

1. *"The min-entropy bound is elementary."* The bound is; the contribution is the separation between native, adapted and bits-back codes. That separation is not expressible in unstructured synonymous coding [1, 2], where a codeword names the set and no walk is traversed. It is also not found in the bits-back literature, where the latent variable is not a route on a shared graph.
2. *"Adapted codes simply code the meaning."* Yes, at rate $H(M)$. But they must be decodable decision by decision on the shared graph, and Theorem 2(b)–(d) shows that this constraint costs decoder memory when the meaning is context-dependent.
3. *"Why not always use bits-back?"* It couples messages, needs an initial overhead of $H(\Pi \mid M)$ bits, and loses the per-message decodability that a walk provides. The three schemes trade shared knowledge, memory and latency.

References

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